

The many applications of Web robots

1. Chatterbots allow people to ask questions and formulate a proper response.
2. Web bots help in handling many tasks such as reporting weather, sports scores, zip-code information, etc.
3. They are used to develop different messengers like SmarterChild on AOL Instant Messenger, etc.
4. They are widely used for Web page scanning and indexing.
5. They can report real-time vehicle locations for the trucking industry.
6. They can also mimic the tasks performed by a psychiatrist, if loaded with many responses.
7. They report air quality.
8. Gaming bots are used for many gaming applications as well.
9. Auction-site Web robots make the process of auctioning easier.

Advantages and disadvantages of bots

Advantages:

1. Help in performing different actions on Web applications at a much faster rate than when the same task is done manually.
2. Help in retrieving the data required from large sets in the least possible time.
3. Can be quite helpful for different site owners to create additional traffic if the site is being included for some index.
4. Web robots are always more reliable and efficient in the actions that they perform as compared to when the same task is done manually.

Disadvantages:

1. There are temporary denials of service, at times, to other visitors reaching a site, which is caused by heavy bursts of page downloading. Visitors may not be able to access site pages at normal speed, or in some extreme cases, not at all, since a robot is busy downloading the pages in a heavy burst and consuming most of the Web server resources.
2. The use of Web robots also results in wastage of bandwidth, as some of the contents that these bots download may not be useful to a person at a later time; hence, the chances of downloading content and then throwing it away is higher. This can ultimately add to bandwidth loss.
3. Different malicious web bots are used by hackers to check for vulnerabilities in the Web system, and they exploit the same for different malicious activities.
4. Malicious Web robots are also used to spam emails of users in order to retrieve user information.

Preventing the problems associated with Web bots

We can prevent some of the issues that arise from using Web bots by taking the following precautionary measures.

1. *Check if we are actually being visited by Web robots often*

The Impact of Bots on Website Security

Good Bots

- Search Engine Crawling
- Website Health Monitoring
- Vulnerability Scanning
- Fetching Content
- Powering APIs



Bad Bots

- DDoS
- Site Scraping
- Comment Spam
- SEO Spam
- Fraud
- Vulnerability scanning

Figure 4: Impact of Web bots on website security (Source: googleimages.com)

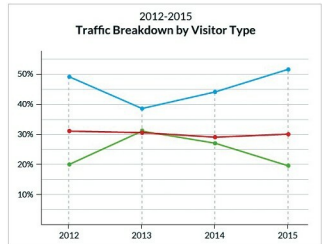


Figure 5: Breakdown of Web traffic on the basis of visitor type during 2012-2015 (Source: googleimages.com)

enough before taking any other measures: We can check if robots are visiting our pages by examining our Web server log files. We can look in the *user agent* field for unusual names, the frequency of their visits and also for the type of files that are being retrieved.

2. *Use the meta tags, robots.txt file and other ways to suggest to robots how your site should be indexed or not indexed:* Try suggesting to the robots as to how you would like your site to be indexed or downloaded.
3. *Convert all time-consuming scripts to the static pages if possible:* It takes longer for any Web server to execute the scripts that generate Web pages than to serve different static HTML pages. Hence, it is less likely that your server would get a denial-of-service when Web robots download static files in heavy bursts.
4. *Check if a possible Web robot requested the script before performing any time-consuming functions:* If you have to use scripts on your website, check if any possible unfriendly Web bot called the script before going into time-consuming parts of the scripts.